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ABSTRACT VISUALIZATION OF WIKIPEDIA NAVIGATION DATA

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Abstract

This work focuses on the analysis of local web navigation without *teleportation* through the design and implementation of a visualization tool. Using data from Wikispeedia as an example of local web navigation, the tool abstracts user navigation behavior into general navigation patterns. By providing interactive visual representations of navigation paths and user decisions, the proposed system focuses on the analysis of how users traverse information networks under the local knowledge constraints of a webpage or article.

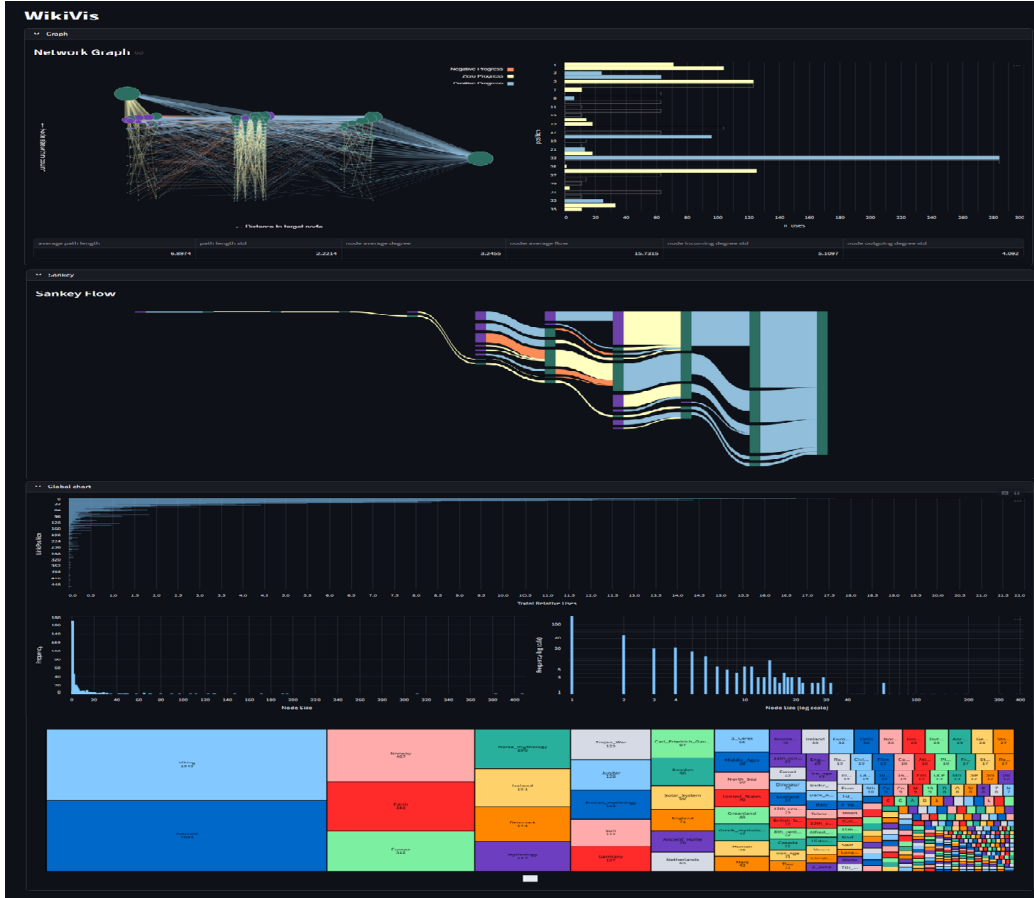


Figure 1: Maximally zoomed out view of the resulting application.

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1 Introduction

In 2021, The company whose name used to be Facebook, was renamed to Meta, with the declared intention to develop a “metaverse” a digital space, “often projected in 3D” [1] that encompasses a large population of users where those users can socialize and use as a second universe. In 2021’s founder’s letter, Mark Zuckerberg claimed it “the next chapter of the internet” [2]. Now, the fact is, that people navigate VR environments differently to the internet, and the current attempts meta has made, are more focused towards generating a new space, instead of advancing the current internet. This has been a source of motivation towards answering one question, if the internet were to be turned into a single, navigable VR space, how would it look like? That question, can be answered with a great variety of opinions, for instance the novel *Ready Player One* [3] fancies a similar idea, however, the fact remains that the navigation of the internet looks nothing like the navigation of a single, navigable VR space, which has extensive literature investigating it [4], and thus evaluating how users navigate the web in a broad scope may be ineffective. If both kinds of navigation are broken down to a simple graph navigation, one easily observable difference between the two is that while navigating internet in a broad scope, the position of the user has diminished relevance. The reason behind this will be referred as *teleportation* based on the terminology that defines a very similar phenomenon in the study of PageRank [5] algorithm, which describes when a user randomly appears in a different node. This event in our scope can be caused by, for example, the use of search engines. Since the study of navigation with *teleportation* would disturb building a single cohesive navigable world, in this study the focus falls into navigation without *teleportation*.

1.1 Wiki Game

“Wiki Game” is a game with a variety of implementations, that consists of, from a pair of Wikipedia articles (s, t) that can be chosen or selected at random, reach t from s only clicking hyperlinks. Some variations push the player to achieve the goal in a timed setting, some count the number of total clicks, and some pit players one against another, turning the game into a direct competition with the winning condition relying on arriving at the goal faster than your opponent. This sets an instance of navigation without *teleportation*.

1.2 Wikispeedia

In 2009, the Data Science & AI Lab [6] from EPFL [7] published in the International Joint Conference on Artificial Intelligence a paper, *Wikispeedia: An Online Game for Inferring Semantic Distances between Concepts* [8] by West, Robert, Joelle Pineau, and Doina Precup. In this paper, a platform to play the Wiki Game was published, along a database of recorded games with the employed educational snapshot of Wikipedia. Thanks to this, there exists a framework to gather data, as well as a copyright licensed database, which contains over 50.000 finished player paths and is available for, among others, academic use. The platform gives the player a subset of randomly generated pairs from which to choose, as well as the option to select your specific pair, if you wish so. Once selected, the platform loads a cleaned version of the selected starting article s . In this cleaned version, all navigation tools are deleted, and only the core article remains, with all available links highlighted and the option to open your target article t in another window in case of not knowing what the name refers to. Upon finalization, Wikispeedia displays a few metrics, and a leader-board. First, it's important to evaluate the platform from which the data was obtained (Wikispeedia) this page has a front page, as seen in figure 2, the in game article page of figure 3, and an end page.

As seen in figures 2 and 3, available links are highlighted, all navigation bars, and external links are removed. An option to surrender is in the "start over" link on top, and players may also report when thinking the current article should link to the specified target.

Regarding the dataset [9], it sums up to three published files, one containing the navigation data, the second containing plaintext version of article content, per article, and a third one containing the full HTML package of the Wikipedia version used. Specific information regarding the used files can be found on appendix A, but a few things worth mentioning are that the navigation package contains the data of finished paths, and the data of unfinished paths separated. In this work, the main focus will be in finished paths, for unfinished path information although interesting needs further evaluation, such as whether or not the player has fallen to a situation where recovery is impossible.

Finally, Wikispeedia allows for backtracking, which is not a common Wiki

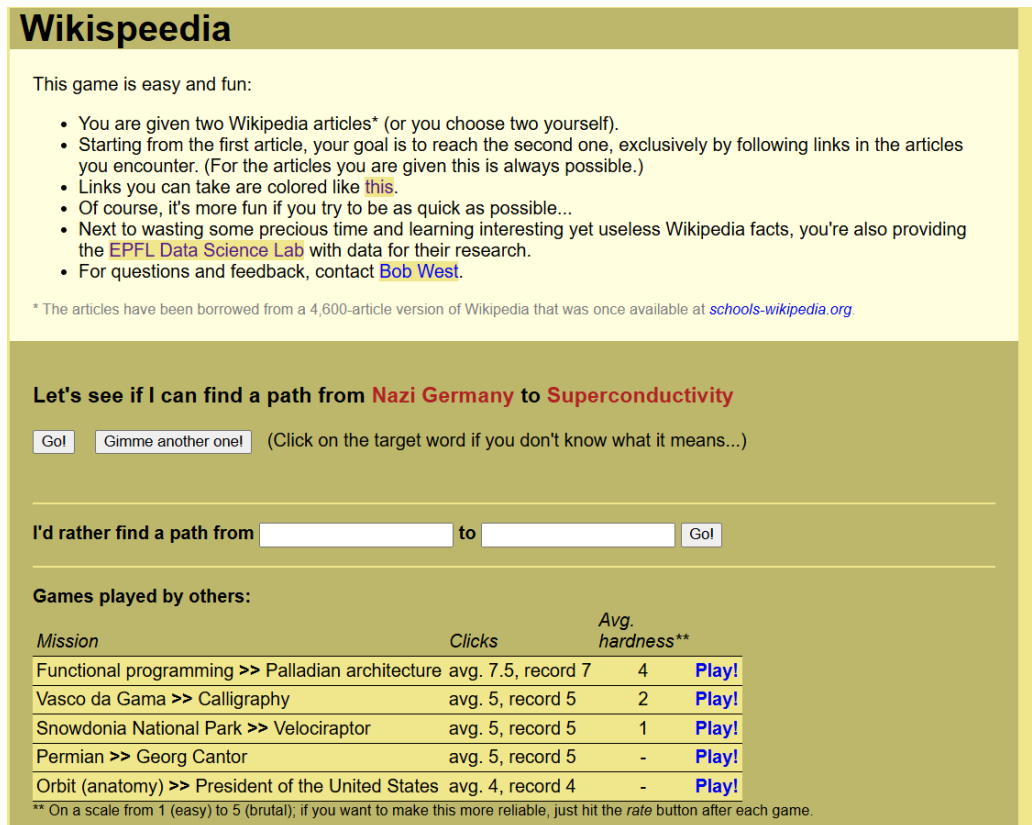


Figure 2: Front page of the online game Wikispeedia.

Game rule, but would be harder to regulate in a free use platform.

2 Similar work

Regarding Wikispeedia, several research works have used this dataset, marking it as a valuable source of information for studying human navigation behaviour, analysing semantic relationships between concepts, extracting meaningful patterns, and developing machine learning models. The original work from which the dataset originates, *Wikispeedia: An Online Game for Inferring Semantic Distances between Concepts* [8] by West et al., introduced the dataset through the analysis of user-generated navigation paths within Wikipedia. The authors used the collected navigation data to infer the

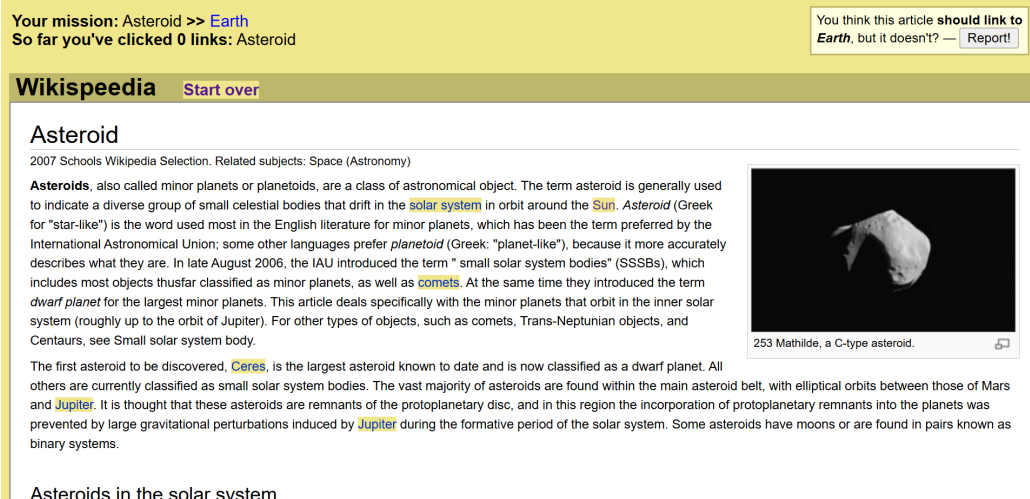


Figure 3: Example of article "Asteroid" as seen from the game platform.

semantic distances between concepts, using the observed paths to validate distances between different concepts and subsequently cluster them according to their semantic similarity. Beyond the initial study, the Wikispeedia dataset has been employed in further research efforts aimed at understanding and predicting user navigation behaviour. For instance, several works have explored whether navigation patterns can provide enough information to determine the success of a user's attempt at reaching a target concept. In *"On the right track! Analysing and predicting navigation success in Wikipedia"* [10], Koopmann et al. investigate the relationship between users' navigation strategies and their ability to successfully complete a given task.

3 Objectives

The primary objective of this work is to analyze user navigation behavior under a no-teleport setting, where users are required to follow the available connections between concepts rather than directly jumping between unrelated nodes, performing an iterative local decision-making in their navigation. To achieve this objective, and taking advantage of the accessibility of the Wikispeedia dataset, the analysis is structured around a series of research questions that aim to investigate different aspects of navigation patterns, user behavior, and the underlying structure of the explored paths. Specifi-

cally, this work addresses the following questions:

- How do users navigate a Wikispeedia game?
- How much do main paths vary within a game?
- Why do users take the paths they take?

Specifically, in this document a game $\mathbf{G}$ is generated by all the finished paths from a start article s to a target article t , and contains the information of all entities that started at s and finished at t . These games are visualized individually upon selection of (s, t) .

3.1 How do users navigate a Wikispeedia game?

The focus of this question is, in a general sense, to understand what mechanisms the players use to reach the goal. For this purpose, it is of interest to define and gauge progress within the game, it is also of interest to see what connections are the most used, the distribution of use of said connections, and anomalies within these connections as well.

Specifically, a series of requirements can be built:

- The system shall be able to capture and reconstruct complete user navigation paths within a Wikispeedia game session.
- The system shall define and compute a metric for player progress toward the target article during gameplay.
- The system shall identify and analyse the articles most frequently used by players.
- The system shall compute the distribution of link usage across the game dataset to understand navigation preferences.
- The system shall detect and report anomalies in navigation patterns, including rarely used or unexpected transitions between articles.
- The system shall provide aggregated statistics describing general navigation behavior across multiple players and sessions.

3.2 How much do main paths vary within a game?

This question, which is very much related to the previous one, focuses on the differences. It is expected that many players will take similar routes, and thus generate flows that may diverge and converge. It is of interest the capacity to visualize the information in a way where these flows are obvious, or at least forward to spot.

Specifically, a series of requirements can be built:

- The system shall identify and extract the main or most common paths followed by players.
- The system shall quantify the variation between navigation paths within the same game instance.
- The system shall measure divergence and convergence points in user navigation flows.
- The system shall support visualization of navigation flows.
- The system shall present navigation data in a way that makes flow patterns and variations easily interpretable.

3.3 Why do users take the paths they take?

Finally, this question focuses on the why. In this case, the experience a player might have playing one common game may strongly vary depending on external factors, such as screen size, browser of choice, connection latency... This means that this question can't be answered in an absolute manner, and some assumptions must be made.

Specifically, a series of requirements can be built:

- The system shall collect contextual information that may influence user navigation behavior, such as article metadata.
- The system shall correlate navigation paths with contextual and environmental variables to identify potential influencing factors.

- The system shall support hypothesis-based analysis, acknowledging that causal inference may not be directly observable from the available data.
- The system shall identify patterns that suggest potential motivations behind user choices, without assuming definitive causality.

4 Implementation

The questions previously mentioned may produce varying answers for specific cases, however web navigation is not homogeneous, and thus individual cases may vary widely. For this reason, instead of answering the mentioned questions outright, a visualization tool has been produced. This tool allows for individual examination of games, and answers the questions by the use of a diverse set of visualization techniques. One important mention is that all paths have been idealized; since Wikispeedia allows for backtracking, but backtracking an edge does not necessarily generate a real edge, all backtracking cycles are cleared and only the final, non-backtracked edge is represented in the final visualizations.

This tool is implemented using Python [11], and the used packages are a Streamlit [12], Streamlit Echarts [13], which is a Streamlit component that visualizes ECharts [14]. Besides these main components, the tool also uses Altair [15] for some auxiliary charts, and more core components such as Pandas [16] and Numpy [17].

4.1 How do users navigate a WikiSpeedia game?

This question poses the problem that navigation can be represented in various ways, and different levels of abstraction, flow diagrams, graphs or trees could work under specific circumstances, so the first step would be to examine the data. After some initial exploratory analysis and reading the documentation on the headers of the dataset, as seen in appendix A, it is concluded the data can be summarized as a directed weighted graph [18] with cycles. Regarding the weights, rather quickly can be seen that highly visited articles are few, meaning there are great deal of smaller flows. For this reason, the best approach to answer the original question was decided to be a graph, this allows to visualize navigation and flows between articles, and is not as affected

by minor flows or cycles as a flow diagram would be. Specifically, it is defined that a directed graph $G = (N, E)$, where N denotes the set of nodes and $E \subseteq N \times N$ denotes the set of directed edges. An edge $(n, m) \in E$, with $n, m \in N$, represents a directed connection from node n to node m . Furthermore the distance d_{nm} is defined as the minimum number of edges in a path from node n to node m . Finally, progress within the game is defined as a transition from node n to node m satisfying

$$d_{nt} > d_{mt},$$

where t denotes the target node. That is, the transition decreases the distance to the target node t . By this definition, all progress $p \in \mathbb{Z}$ achieved by a single edge will be either $p = 0$, $p = 1$ or $p > 0$.

With this in mind, the information encoded in this graph is:

- Distance to t , as the approximate position of a node in the horizontal axis.
- Progress, as the color of the edges in the graph.
- Number of visits per article, as the size of the node.
- Number of uses of a certain edge, as the size of the edge.
- Node type, whether the node n is at $d_{ns} = 1$ or not as the color of the node.

Leaving this graph in its original, unsorted form would result in a highly complex and potentially unintelligible representation. The large number of connections between nodes, combined with the varying degrees of connectivity among them, causes the visualization to suffer from what is commonly referred to as a *Graph Hairball*. This phenomenon occurs when the density of edges and overlapping relationships becomes so high that the resulting visualization turns into an entangled structure, making it difficult to distinguish meaningful patterns, identify relevant structures, or extract useful insights through visual inspection alone. An example of this effect can be observed in figure 4.

For this reason, sorting the graph for the sake of clarity proves to be a must. In this case, the nodes are sorted in the vertical axis by number of visits,

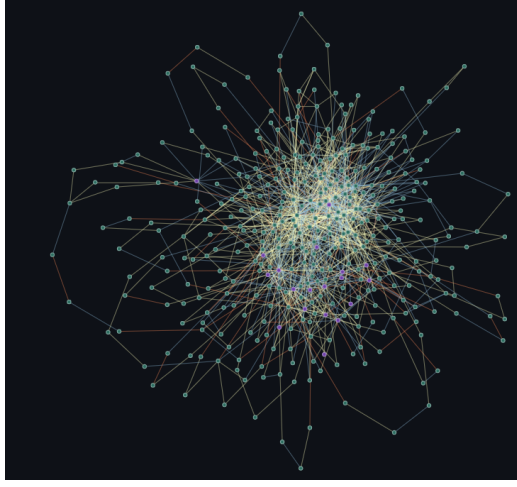


Figure 4: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, unsorted.

pushing most visited articles towards the top of the graph. With a sorted graph as the current one, and the fact that positive progress by a single edge is limited to 1, a grid graph could be proposed. This proposition would highlight edges with non-zero progress, at the cost of overlapping edges of negative progress and zero progress. By an initial data evaluation the specific case of negative progress by a distance lesser than -1 is rare. Furthermore, in the visualized cases, the amount of nodes with distances halfway between d_{st} and 0 is notably greater than those near the beginning or end, meaning that a standard grid would have big areas of empty space, as seen in figure 5.

Instead of a drawing the graph in a pure grid, the columns of the chosen graph expand and collapse to fit the vertical space, this however, difficults the ability to spot how many nodes are in a column at a glance. To fix this issue, a sinusoidal offset is added in the horizontal positioning of nodes, which serves a dual purpose. On one hand, it enhances visibility of edges with 0 progress, by breaking the alignment of the nodes. On the other hand, the offset encodes the amount of nodes by full revolutions, achieving a half revolution every 5 nodes, or a full revolution every 10. This eases the previous problem of not knowing how many nodes there are in a column at a glance and is further supported by auxiliary labels, which can be toggled. By the

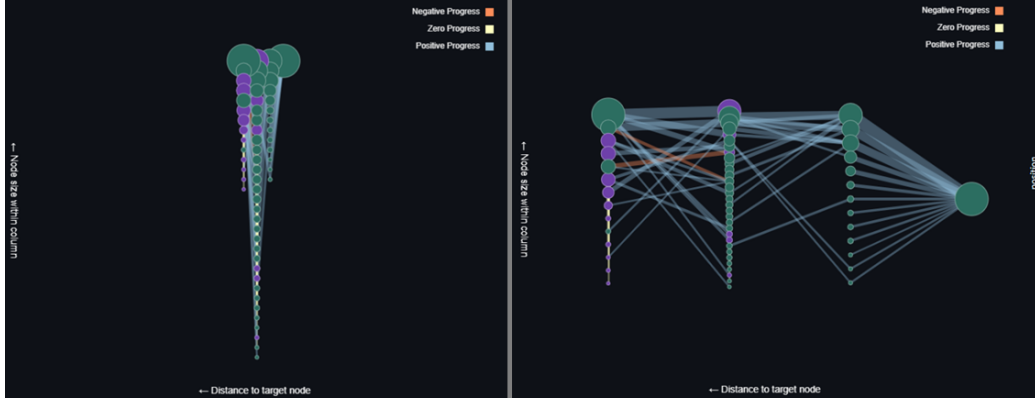


Figure 5: Graphs from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with an edge filter of 1%, with grid aligned nodes (left) and collapsed columns (right).

defined parameters, the final node t will always lie by itself in the final column (since it is the only node with distance 0 to itself), however, the starting node s will lie in a variably crowded column, so to highlight it from the rest, s is added with a significant offset in the vertical axis, making t and it's edges more obvious to the eye as seen in figure 6.

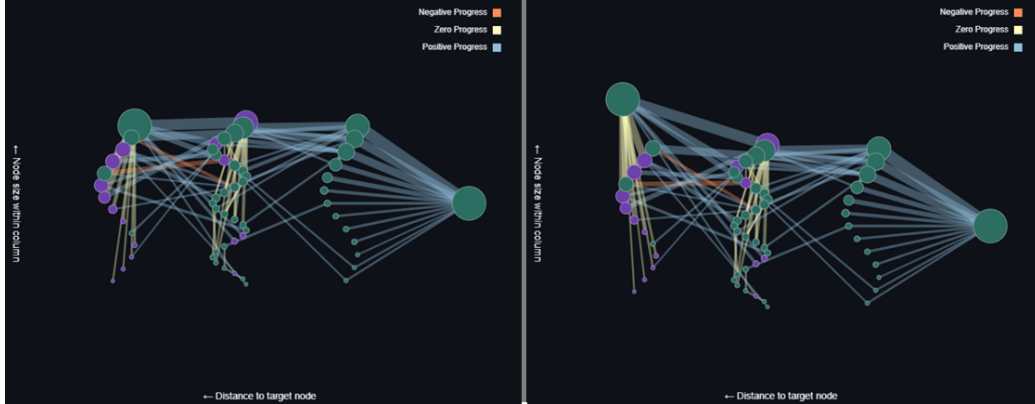


Figure 6: Graphs from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with an edge filter of 1%, with sinusoidal offset, and without Y axis offset on s (left) and with Y axis offset on s (right).

With the current parameters, some information can be disclosed, but there are still high levels of cluttering. For this reason, a filter is implemented, that allows the graph to be filtered by % of uses, filtering the edges that have been accessed by less than a user set percentage of people, selected with a slider. Due to the distribution of the weights, no game $\mathbf{G}$ is found in which a single edge is traversed by 50% or more of the total amount of players, for this reason, the slider value ranges $[0,50]$. Also, the amount of edges with low percentages are abundant, and for this reason the step of the slider is set to 0.5, instead of the default 1.0. This same distribution is the cause of a problem in the encoding of the number of visits per article, which when encoded on a lineal scale, some nodes heavily occlude nearby ones, as seen in figure 7. A logarithmic encoding may help with this, but at that would heavily reduce the visual indication of which nodes are more visited. For this reason the final scale used is a normal logarithm squared. A comparison of these functions can be seen in appendix B. Finally, to de-clutter the final visualization, the size of the edges is divided to represent half the size the nodes do. Finally, some navigation metrics are added as a complement below the graph, specifically information regarding player path length, and incoming/outgoing flows.

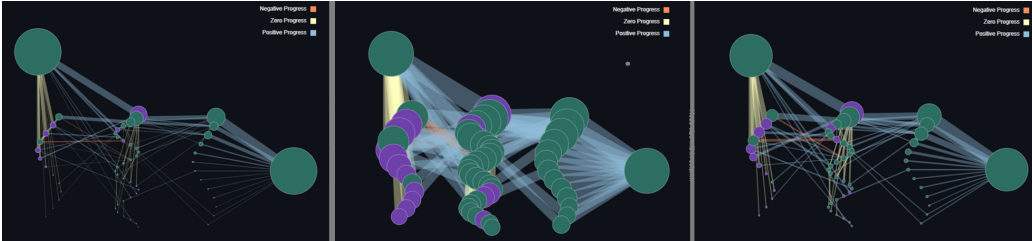


Figure 7: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with an edge filter of 1% with three different size encodings. $x/10$ (left) , $14.47 * \ln(1 + x)$ (middle) and $0.04 * \ln^4(1 + x)$ (right).

4.2 How much do main paths vary within a game?

The abstraction of the graph to ask the previous question has the less than ideal effect of abstracting paths to the sum of their segments, it becomes hard to distinguish individual tracks and some relevant metrics such as path lengths, or path causality. For this reason, to answer how much do main

paths vary within a game, another method proves more useful. Specifically, to display the comparison within flows, a different method is proposed.

Sankey diagrams [19] are visualization tools commonly employed to represent flows, the way a Sankey diagram encodes the flow into the width of the arrows is reminiscent of the width of the edges in our graph, and maintains the cohesion of the encoded information, making a Sankey diagram the used visualization to represent variation from diverse flows.

The process to decide the flows to be visualized is based on the first jump of the initial node, by isolating the nodes n with distance to s $d_{ns} = 1$, afterwards the edge with highest amount of uses is followed, until the final node is reached. By this process, the main direct flows can be extracted, and so can be the merging points. Afterwards, this flows are traced back, to set the distance of every node in a flow to the final destination. This flow distance is used to position the nodes in the horizontal axis. To be precise, our Sankey diagram encodes:

- Node type, whether the node n is at $d_{ns} = 1$ or not. As the color of the node.
- Progress, as the color of the edges in the graph.
- Number of visits per article, as the size of the node.
- Number of uses of a certain edge, as the size of the edge.

Regarding the size encoding, edges are represented linearly, since the main focus is on major flows, the scenario where many little flows clutter the view, or are occluded by major flows, is heavily reduced, and thus not a major problem anymore. With this said, it can still become a problem when there are many minor flows that stem directly from the original node, for this reason, another slider is implemented, that allows the user to select the amount of flows to display at once, the program sorts the flows by size of the original node, which equals the starting size of the flow. Since only major flows are displayed, all the smaller flows are not accounted for in the size of the node, this may give false impressions of node size where a highly visited node's major flow is a low percentage of its visits. This issue is corrected on selection, explained below at section 4.4.

The comparison between multiple flows is displayed in figure 8, where it can be seen that the main problem to visualization is not the width of the edges, but the fact that when one flow encompasses too many nodes, all paths collapse horizontally.

This later problem is also solved by the slider, as the observed length of main paths seems to be close to the minimum distance, this can be evaluated from the data, which can be seen in appendix C. The nodes are sorted by the relative size of the edges where they flow into. To be precise, the diagram is constructed by appending iteratively the most used edge from all nodes n with distance to s $d_{ns} = 1$ until t is reached, and then they are sorted vertically by their respective merger node, and then by the flow volume of the edge connecting them to said node. This guarantees that the flows will never cross, and pushes flows with the highest volumes to the top right corner.

Finally, the Sankey diagram is connected to the graph slider, and all edges and nodes that would be filtered out by the slider, have their opacity reduced, to signify that those flows are less than the amount marked by the slider as well.

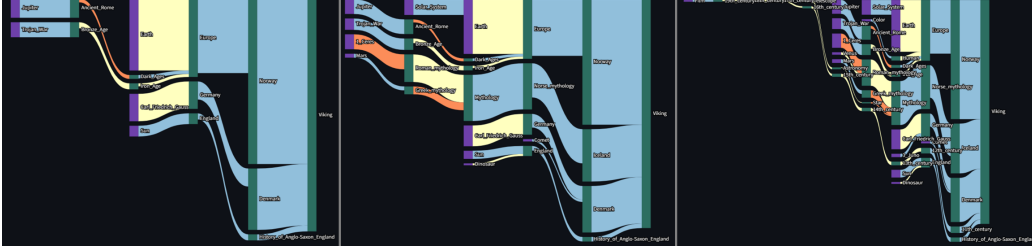


Figure 8: Sankey diagram from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, displaying left to right 5, 10 and 15 flows.

4.3 Why do users take the paths they take?

As mentioned in the objectives section 3.3, completely answering the question is impossible with the available data, so some simplifications are performed. Instead of assuming the exact positions of the links, each article is parsed, and all links are extracted as they appear along the text. This gives a fair approximation of which nodes appear sooner.

To represent the use of links, a bar chart is proposed, which encodes the number of clicks and the position of the link along the text. After visualization of this information, one problem was apparent; the data contains from which node n to which node m a player traveled as seen in appendix A, but it does not specify which specific link was used. In many cases, there are more than one hyperlink that go from n to m in one article, and thus there is no guarantee on which hyperlink the player used. To solve this, and thanks to some small testing with relatives and friends, which is further specified in appendix D, a relevant observation was made, which is that Wikispeedia specifically highlights all available links, as seen in figure 3, this causes players to not read around the hyperlinks, and only read the highlighted content. For this reason, an assumption falls in place; when multiple hyperlinks point to the same article, players will often use the first one.

Based on this assumption, the method used to mark these hyperlinks as repeated, is hollowing the bar. This marks the fact that the link is repeated in a visually obvious way, and often is easily seen which hyperlink it repeats, since it still shares its length. However, when dealing with low number of clicks, the fact that the bars are hollow can be lost, for this reason, bars are hollowed and greyed out. The bar chart, is specifically a horizontal bar chart, to mimic the verticality of a web page, and encodes:

- Position in the text, in the vertical axis Up-Down in ascending order.
- Progress, as the color of the bars in the chart.
- Number of uses of a certain edge, as the length of the bar.
- Node type, whether the node n is at $d_{ns} = 1$ or not, as the color of the node.

With this information, a horizontal bar chart as seen in figure 9 is produced per every node n within a game $\mathbf{G}$.

The game with most nodes contains roughly 400 nodes, which mean roughly 400 horizontal bar charts. Since that kind of information is not one to easily visualize at a glance, these charts are generated dynamically on demand, with the selection methods explained below in section 4.4.

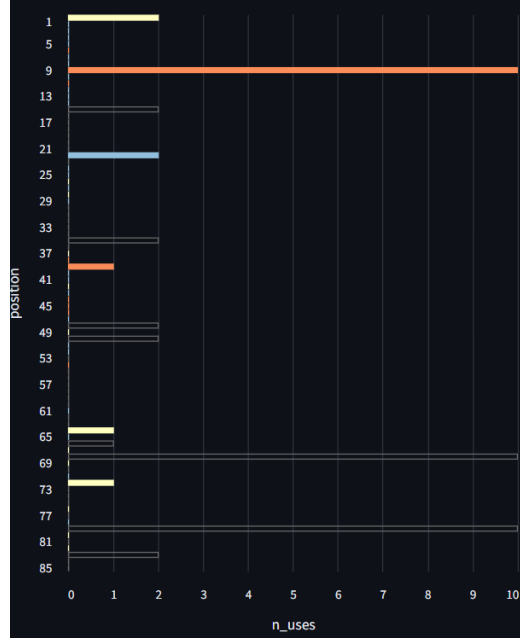


Figure 9: Chart of node religion within game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"}, \text{"Viking"})$ with gray, hollow bars.

To maintain position within the visualization, which enhances the usability of the tool, the chart of node s is displayed by default, and substituted by the appropriate chart on node selection. Besides this local information, the sum of the normalized charts are added to a single global cumulative hyperlink position chart, that displays user tendencies regardless of node information. This chart is further described in section 4.5.

4.4 Detail on demand

All the information visualized in this work up to this point is mostly generalistic, and among these volumes of visualized information, it's easy to lose the specific information contained within each unit. For this reason, the tool possesses different selection mechanisms, with different methods to hide or display information on demand. These tools are located in a collapsible lateral container titled "Controls" as seen in figure 10, that can be opened by toggling a button located in the top left of the screen. The most notorious examples of the ones mentioned previously, would be the sliders that allow

the user to select, the minimum percentage of people who used every edge in the graph, and the one that selects the number of flows displayed in the Sankey diagram.

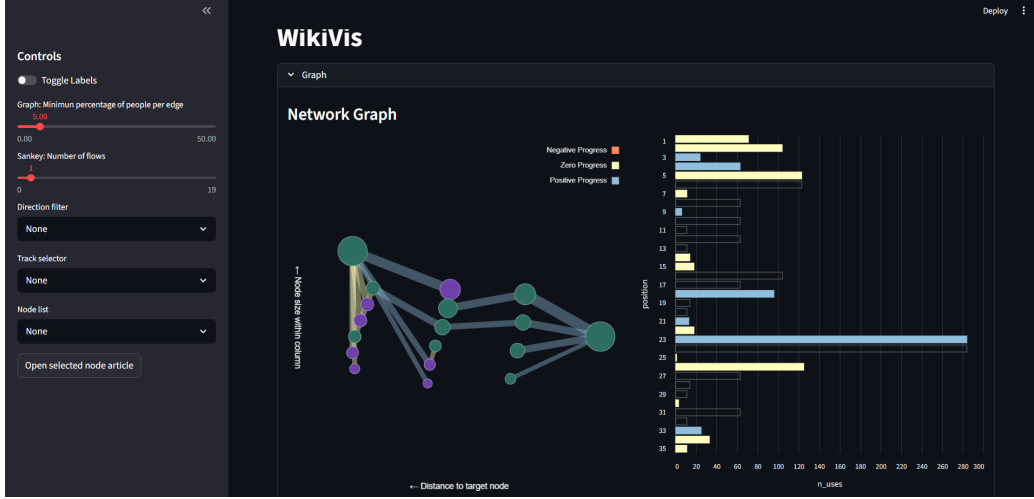


Figure 10: Selection filters (on the left), and extra details on demand tools, with the default values.

One notable design decision in the visualization is that node labels are disabled by default. This choice was motivated by the density of edges in the Graph visualization, where there is high levels of cluttering when displaying high amounts of nodes. For this reason, the label switch that lies on the top right of the panel is turned off by default. This same switch also controls the display of the support labels that mark the amount of nodes per column, as seen in figure 11.

This decision may make labels more prone to cluttering the visualization, so to avoid labels cluttering one another, a consistent label placement strategy was adopted; node name labels are always positioned to the left of the node they describe, whereas the column-count labels are always positioned to the right of their corresponding nodes. This separation reduces the likelihood of overlap between the two categories of labels and helps preserve the legibility of the visualization. Furthermore, placing the column-count labels on the right aligns them with the outer side of the peaks generated by the sinusoidal node distribution. As a consequence, these labels occupy regions of

the visualization that are comparatively less crowded, allowing them to convey additional information while contributing only minimally to the overall visual clutter. Furthermore, column-count labels have partial transparency, to maintain the ability to see edges through them.

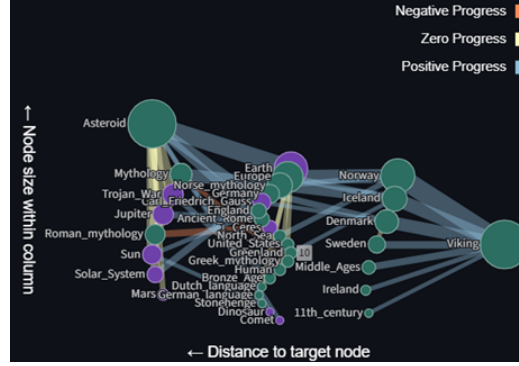


Figure 11: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"}, \text{"Viking"})$, with an edge filter of 2% with labels enabled.

Below the labels and the sliders, specific selection mechanisms are positioned. The first mechanism is a slider that allows the user to filter out all the edges except those with a specific progress (Negative Progress, Zero Progress, Positive Progress). A comparison of this filters is visualized in figure 12. This filter allows for the user to spot nodes where players went against the intention of the game, this can mean the node is poorly connected in the context of one specific game $\mathbf{G}$, or it may be a point of confusion due to common misconceptions, both cases of interest when analyzing the structure of the web.

The next item in the controls is a track selector, this allows the user to select from the complete list of tracks, which are sorted by descending track length, meaning the tracks near the top of the list are the shortest paths, and the ones near the end are the longest paths within the game $\mathbf{G}$. This allows to discern individual recorded paths from the abstractions, and even compare from same length paths. Hovering the track on the list, displays the full track names, for an easily readable list.

Upon selection, all other nodes and edges visibility is reduced, and the se-

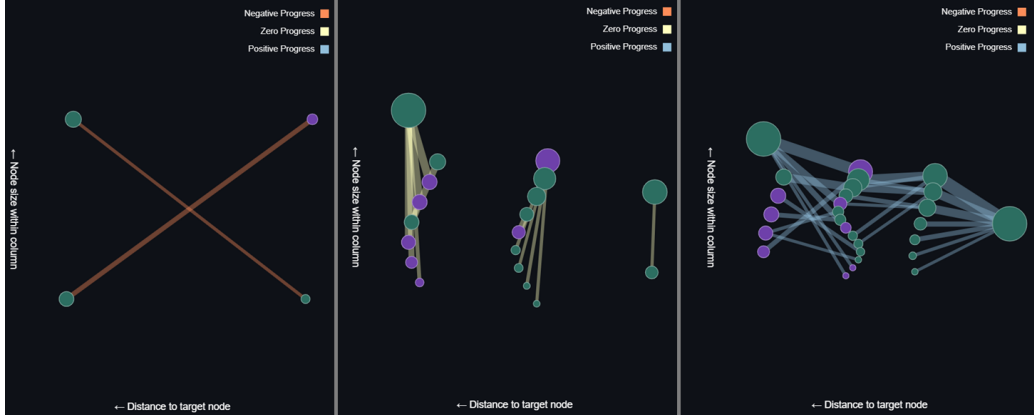


Figure 12: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with an edge filter of 2% with three different direction filter settings, from left to right Negative Progress, Zero progress, Positive Progress.

lected ones are shadowed with a magenta hue. This shadowing also applies to labels, making labels of selected nodes distinct even when general labeling is on, and even when labeling is off, the labels of the node still display, since even on long paths, the sliders and other filters work simultaneously as means to reduce the hypothetical clutter. A comparison of track selection and label interaction can be seen in figure 13.

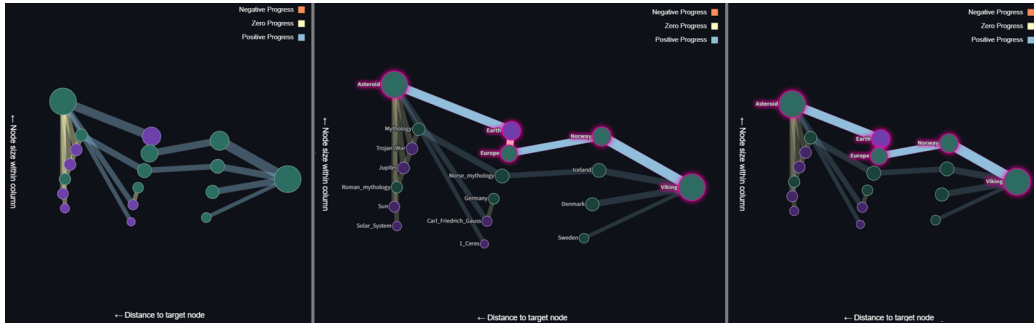


Figure 13: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with an edge filter of 5% and selection of path ["Asteroid", "Earth", "Europe", "Norway", "Viking"] comparing left to right; No selection, Selection of path with labels, Selection of path without labels.

The next tool is a node selection list, this allows to look for specific nodes by name. The list is sorted by number of visits to the node, and upon selecting a node, the specified node is highlighted, as well as all directly connected nodes. This selection, also changes the chart on its right to be the specific chart of the selection node. As in the track selection case, node selection also shadows nodes, edges and labels with a magenta hue.

When selecting a specific node, the Sankey diagram completely shifts from general flows, to specific node flows, where the selected node is set at the center of the diagram, given the selected node n_s and any connected node n all incoming edges where $E = (n, n_s)$ are set to the left of the node, and all the outgoing edges such that $E = (n_s, n)$ are set to the right of the node. The edges are vertically sorted by flow volume. This specifications of the Sankey diagram make a special circumstance, which is the case when the chain of incoming and outgoing nodes form a cycle. In this case, when there's a bidirectional connection between nodes n_s and m such that $E = (n_s, m)$ and $E = (m, n_s)$ both exist at the same time, the node m labels on the right change to the selection color, to highlight that the node displayed is a copy of an existing node, and thus the same node appears twice. This can be seen in figure 14. It is also worth mentioning that since node selection changes the Sankey diagram to visualize all flows of one specific node, the size mismatch mentioned in section 4.2 is naturally corrected, and the of the selected node becomes the complete size.

Related to the selection mechanisms, the node selection method is extended to the Graph, Sankey diagram and Bar chart, meaning that clicking on a node in any of the previously mentioned, will select that one specific node. As clearing selection of a node may be as useful as selecting a node, a mechanism is implemented where selecting one same node twice cancels the selection, selecting a track also cancels the selection, and all filters work parallel to selection, with the exception of the slider controlling Sankey diagram flow amounts, since flow visualization hides non-selected flows.

Finally, as seen in figure 10 the last item in the "controls" container gives the user the option to open the selection node. As mentioned previously, the accurate way a player may experience the game can only be approximated, by opening the specific node in the browser gives a rough idea of the link distribution, information, can open up to guess the reasoning behind player

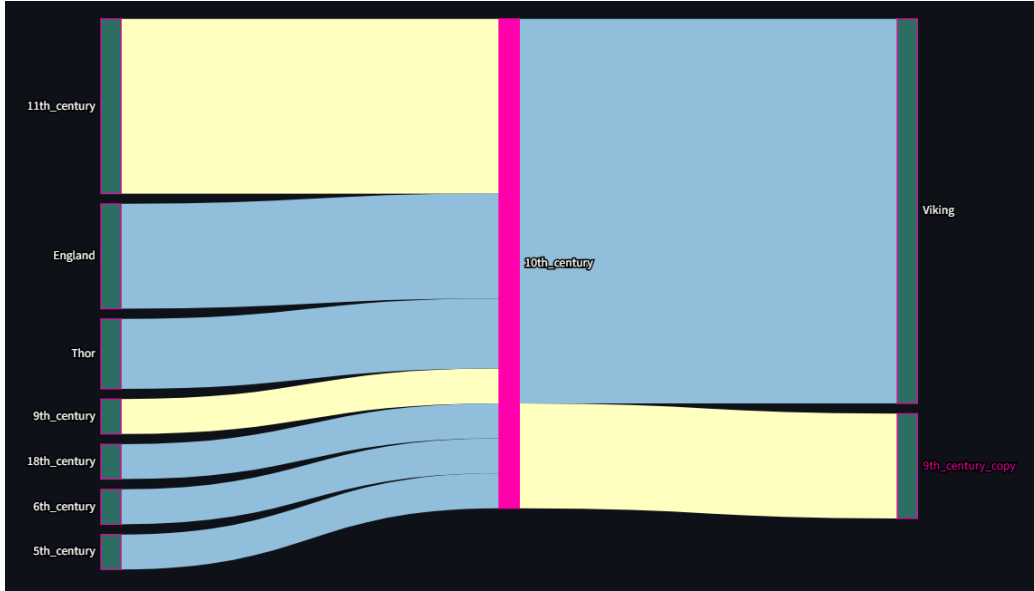


Figure 14: Graph from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, with a cycle marked by label color (bottom right).

decisions.

4.5 Global Metrics

To complement the information given by the previous visualizations, and extract general insights of a specific game $\mathbf{G}$ an extra container is generated. The first item in this container, is the previously mentioned global cumulative hyperlink position chart, which can be displayed on figure 15. This chart sums the most used hyperlink positions in the game $\mathbf{G}$, which allows for general insights, instead of those pertaining to specific nodes. Following the reasoning behind the chart local counter-part, this chart also is a horizontal bar chart.

To evaluate according to multiple sources, instead of relying only on the most visited nodes, each graph chart clicks are normalized before adding. This visualization has one specific problem: when there's high variance between the length of the hyperlinks lists, some nodes of low lengths may contribute only to highly placed positions, and this could sum up. However the alternative

(normalizing positions) would imply the fact that positioning of hyperlinks among nodes is consistent, which is not factual.

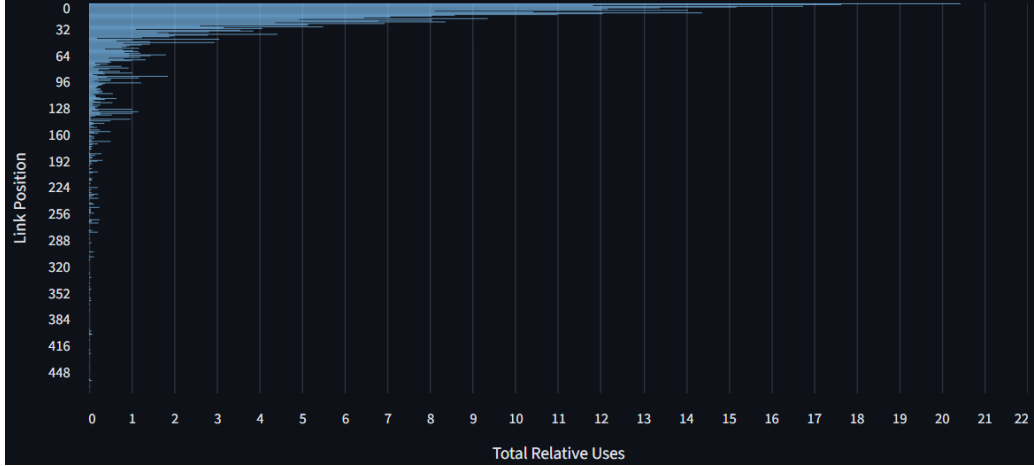


Figure 15: Global cumulative hyperlink position chart, which contains the sum of the normalization of the positions of all links within the game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$.

Other useful information would be the distribution of the article visits, which can be represented by a histogram. By observing the resulting distribution, similarities to power law distribution are found, and for this reason a secondary chart is drawn, one that encodes the same information but is in a log-log scale, as seen in figure 16. It is worth mentioning that the starting and target nodes s, t are removed from the plot, since each of those node sizes encodes the total amount of people.

On a final note, it is still partially unresolved which nodes are the ones that are most visited in all the game $\mathbf{G}$, since previous visualizations only answer that question conditionally, such as the graph visualization, which answers which nodes n are the greatest per each distinct distance d_{nt} , or the Sankey diagram which by encoding flows, displays edge size instead of node size. For this reason, one last visualization is implemented, which consists of a Treemap [20] encoding all nodes and its sizes, as seen in figure 17.

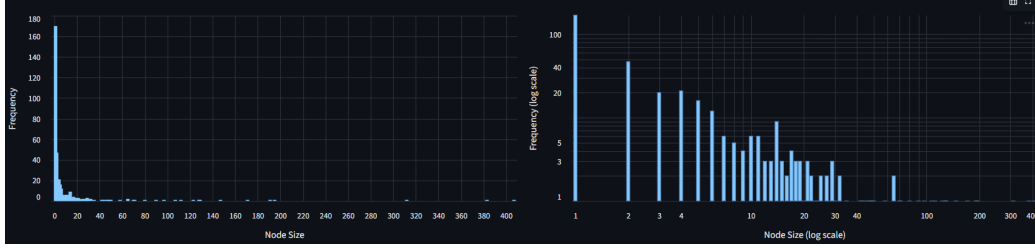


Figure 16: Charts from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$ representing the frequency of node sizes, with linear scale (left) and log-log scale (right).

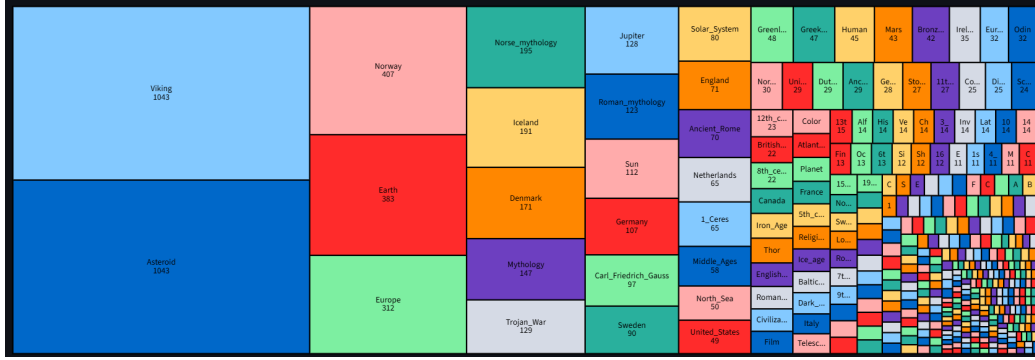


Figure 17: Treemap from game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, encoding the size of each node.

4.6 Specifications note

To improve both performance and usability during development, the application makes use of the caching mechanisms provided by Streamlit, as well as the data management capabilities of Pandas. A significant portion of the information generated and processed by the system is either cached in memory or persisted to external `.csv` files. By storing intermediate and final computation results, the application avoids repeatedly performing expensive data-processing operations each time it is launched.

This approach provided considerable advantages during the development and debugging stages. In particular, it enabled the dynamic modification of the selected game, denoted as $\mathbf{G}$, without requiring the complete graph structure and all associated metrics to be recomputed from scratch whenever the ap-

plication is reloaded. As a result, experimentation with different subsets of the dataset becomes significantly faster, allowing for a more efficient iterative development process and reducing the waiting time associated with repeated executions. An additional benefit of persisting graph-related data to `.csv` files is the ease with which generated datasets can be exported, archived, and shared with other researchers or developers. This facilitates reproducibility, enables external analysis using third-party tools, and simplifies the exchange of results between collaborators.

The platform itself was implemented and evaluated using Chromium-based web browsers. Throughout development and testing, these browsers served as the primary execution environment, ensuring consistent behaviour of the user interface and visualization components. While the application is expected to function correctly on other modern browsers, formal testing was conducted exclusively within the Chromium ecosystem.

Regarding color, the palette used was chosen to prioritize differentiation among individual element components (among edges, for instance), for this reason, on some occasions different elements such as nodes or vertices may have colors with slightly less contrast. From this palette, the main components that should be obvious to the eye have been picked with color blind people in mind, and have been contrast checked, as can be seen in figure 18, where the main colors used can be compared, along with estimated versions given the main types of color blindness. Finally, the general theme was set to dark (dark backgrounds, clear fonts) to maximize contrast with clearer colors, while reducing visual strain.



Figure 18: Color palette, encoding from left to right (Edges/Bars with negative progress, Edges/Bars with zero progress, Edges/Bars with positive progress, Selection color, Nodes n with distance $d_{ns} = 1$) with approximations to the equivalence seen by the three main types of color blindness.

5 Results

As a result of the previous work, a complete functional platform has been developed. This platform allows for individual examination of games $\mathbf{G}$, and for the extraction of insight. These insights, relative to the original questions are described as follows.

5.1 How do users navigate a Wikispeedia game?

On a personal level, an answer to the question "How do users navigate a Wikispeedia game?" is; more efficiently than expected: in almost all the explored games $\mathbf{G}$ the average path length was roughly over 50% of the minimum distance. One insight that was more expected, is that many nodes act as hubs, both by grouping people from distinct nodes, or by dividing into many main flows. This property can even be measured, the higher the standard deviation of the degree of the nodes of the graph, the greater the amount of hubs, and since the visualization is a directed graph, the standard deviation from incoming edges may vary in respect to the standard deviation from outgoing edges. By measuring this difference, it is possible to know if a specific game has a more even expansion into different nodes, or a more even collapse into the final node. By comparing the values among top 10 most played games $\mathbf{G}$, generally games tend to expand more harshly, and

group more evenly into the final node, this can be confirmed by comparing the metrics of every graph, as seen in table 1. Besides this, the distribution of nodes usage seems to follow a power law, meaning that very few nodes and paths encompass most of the flow, but many different small flows exist.

Game	Avg path length	Path length std	Node avg deg	Node avg flow	Node in deg std	Node out deg std	Num of players
"Asteroid", "Viking"	6.8974	2.2214	3.2455	15.7314	5.1096	4.0920	1043
"Brain", "Telephone"	6.4346	2.4903	2.5929	16.6725	3.4093	3.1349	1040
"Theatre", "Zebra"	7.0276	2.9193	2.6894	13.3374	5.3294	3.6787	905
"Pyramid", "Bean"	7.1526	2.7976	2.4317	12.5396	3.5039	2.9341	642
"Batman", "Wood"	6.6283	2.7436	1.6392	5.27215	1.9392	1.6825	148
"Bird", "Great white shark"	5.4782	1.9602	1.6590	14.0454	1.5839	1.6130	138
"Batman", "The Holocaust"	5.1344	1.0244	2.1346	9.46153	2.7226	1.8997	119
"Bird", "Adolf Hitler"	5.2523	1.2138	1.9753	5.61728	2.7063	2.5640	107
"Beer", "Sun"	4.9393	1.6892	1.7816	4.48275	2.3841	1.9614	99
"Batman", "Banana"	6.7971	2.1732	1.7448	4.08163	1.8573	1.5684	69

Table 1: Table with the results of the 10 most played games **G** in Wikispeedia, highlighted in green wether the incoming or outgoing degree standard deviation is greater.

Finally, as per the initial requirements:

- The system shall be able to capture and reconstruct complete user navigation paths within a Wikispeedia game session.

This requirement is fulfilled by the track selection tool.

- The system shall define and compute a metric for player progress toward the target article during gameplay.

Progress is defined specifically as a metric, encoded in edge color in both graph and Sankey diagram, and in bar color in the node specific bar chart.

- The system shall identify and analyse the articles most frequently used by players.

The graph visualization sorts vertically by most frequently used articles, and thus enhances their visibility.

- The system shall compute the distribution of link usage across the game dataset to understand navigation preferences.

This requirement is fulfilled by both graph and Sankey diagram vi-

sualizations, and specifically with the edge usage filter.

- The system shall detect and report anomalies in navigation patterns, including rarely used or unexpected transitions between articles.

This requirement is fulfilled by the graph direction filter, in which edges with negative or zero progress can be singled out.

- The system shall provide aggregated statistics describing general navigation behavior across multiple players and sessions.

This requirement is fulfilled by the game metrics displayed below the graph visualization, which table 1 summarizes.

5.2 How much do main paths vary within a game?

Regarding the question "How much do main paths vary within a game?", the fact is, that many of the flows initially divide, but end collapsing into bigger flows, and most of them do so at a rather even pace. One example would be the most played game, the game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$, where from the 19 flows represented in the Sankey diagram, only one doesn't merge into any other, and all other 18 merge into 4 flows before reaching the end. Another example would be the second most played game, in which from 33 different initial flows, only one doesn't merge, and the remaining 32 merge into 5 flows before finishing, as seen in figure 19. It is worth mentioning, that this perspective will always yield more values for the starting node, since the direct jumps from s are the nodes that seed the flows, but this conclusion that most flows collapse into other flows before the end is also supported by looking at the nodes with the tools specified in section 4.4, which also yield a greater degree for s than the degree of t .

Specifically, as represented at figure 20, it can be seen that 33 different nodes stem from "Brain", the node s , while only 14 connect directly to "Telephone", the node t . While this is indicative of trend, the trend is not as strong in the rest of the nodes, for the difference between the mean standard deviation of incoming nodes and the mean standard deviation of outgoing nodes is relatively small $(3.4093 - 3.1349) = 0.2744$. The most clear example of the trend, would be the third most played game, the one defined by $(s, t) = (\text{"Theatre"} , \text{"Zebra"})$, in which 16 initial flows collapse into two final flows

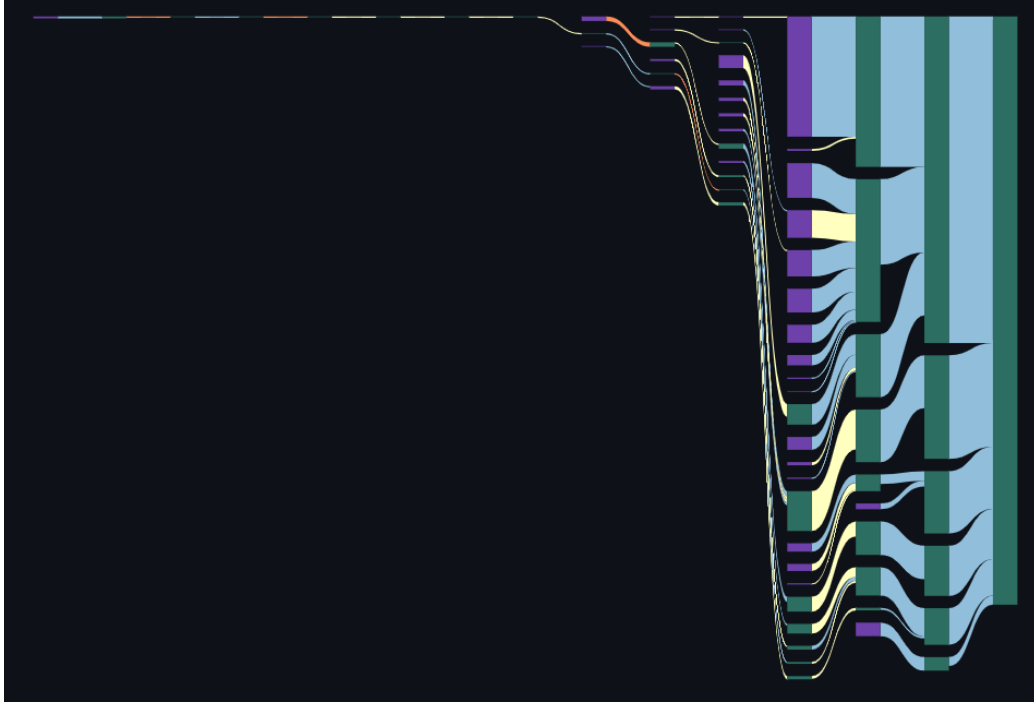


Figure 19: Sankey diagram from game $\mathbf{G}$ defined by $(s, t) = (\text{"Brain"} , \text{"Telephone"})$, without labels or filters.

in the Sankey diagram representation, and when looking at the selection methods, it can be extracted that from the node s 16 different flows stem, while only 7 collapse into node t , and the difference in the standard deviation of their degrees is the highest among the top ranked, specifically of $(5.3294 - 3.6787) = 1.6507$. Also worth noting from figure 20, here starting node s has preceding nodes, these nodes come from tracks that cycle back to the original node. This scenario can't happen with the target node t because once the player reaches t the game automatically ends.

Regarding the only game $\mathbf{G}$ where the incoming degree standard deviation is higher than the outgoing, an anomaly occurs there, in which not one, but two nodes have more visits than the total amount of players, as can be seen in figure 21. This game, defined by (Bird , Great White Shark), has the nodes "Fish" and "Great Whale Shark" revisited often by players, and most movement occurs revolving the "Fish" node. In fact, all original flows in the

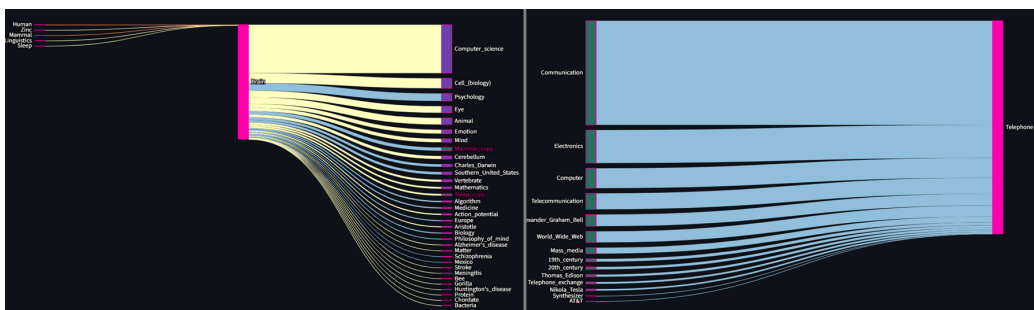


Figure 20: Sankey diagram from game **G** defined by $(s, t) = (\text{"Brain"} , \text{"Telephone"})$, with focus on "Brain" (left) and "Telephone" (right).

Sankey diagram visualization collapse onto a single node, "Fish", which then leads to "Great Whale Shark", which finally leads to "Great White Shark". It is also worth mentioning, that from this game has a total of recorded tracks of roughly 10% of the games with most data recorded, meaning that the whole graph can be displayed, unfiltered, and with labels on, and produce a mostly readable graph as can be seen in appendix C, meaning that to infer powerful statistical value a greater population may be needed.

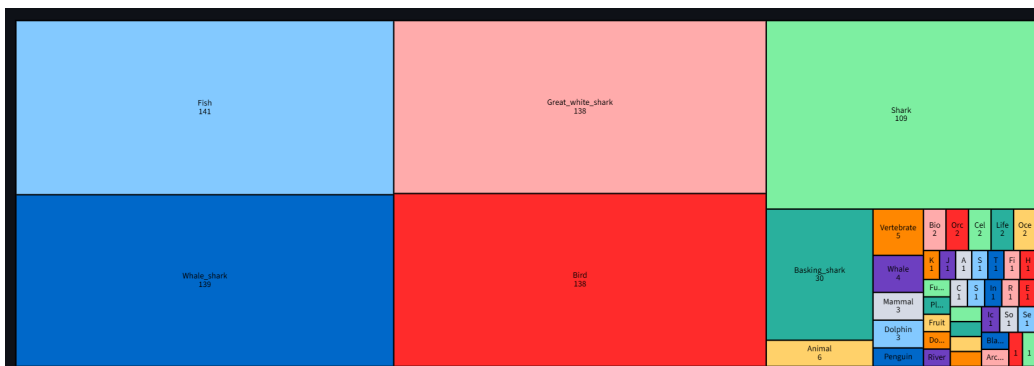


Figure 21: Treemap of nodes of game **G** defined by $(s, t) = (\text{Bird}, \text{Great White Shark})$.

Finally, as per the initial requirements:

- The system shall identify and extract the main or most common paths followed by players.

This requirement is fulfilled by multiple factors, the Sankey diagram visualization gives an approximated summary, which is specifically enhanced with the graph visualization with the edge filter tools.

- The system shall quantify the variation between navigation paths within the same game instance.

This requirement is fulfilled by the conjunction of the Sankey diagram visualization with the metrics below the graph visualization.

- The system shall measure divergence and convergence points in user navigation flows.

This divergence is measured by the different standard deviations of node degrees, and can be specifically delved into with the node selection tool.

- The system shall support visualization of navigation flows.

This requirement is specifically resolved by the Sankey diagram visualization.

- The system shall present navigation data in a way that makes flow patterns and variations easily interpretable.

This requirement is specifically resolved by the Sankey diagram visualization.

5.3 Why do users take the paths they take?

The question of why users take the paths they take during navigation does not admit a single, straightforward answer. Instead, the visualized behavior appears to be the result of multiple factors, each contributing to the track. The most impactful of these factors would most likely be the semantic component. The influence of semantic proximity on user navigation has already been documented in previous research and constitutes one of the central assumptions of the original Wikispeedia study by West et al. which introduced and published the dataset used throughout this thesis, focusing on the concept that navigation behaviour can be used to perform semantic inference.

The semantic component is not limited to the connection between the starting node s and target node t of a game. To be specific, semantic relations are formed in the selection of intermediate nodes as users progressively approach their destination, based on the local decision-making that must be made on each node. This phenomenon can be observed multiple times in the data. Consider the game $\mathbf{G}$ defined by $(s, t) = (\text{"Asteroid"} , \text{"Viking"})$. In this game $\mathbf{G}$, one of the flows visualized in the Sankey diagram navigates through a sequence of articles corresponding to up to 9 successive historical centuries, effectively traversing a chronological chain before eventually reaching t , "Viking".

A second example can be found in the transitions between mythology-related articles. A particularly common route involves moving from node *Roman mythology* to node *Mythology*, and subsequently to node *Norse mythology*. Although relatively short, this path is traversed by a significant number of users and exemplifies how individuals often rely on higher-level conceptual categories as means for local navigation.

Beyond semantic components, the results also suggest a different factor influencing user behavior; While the visualizations and analytical methods employed in this work necessarily introduce certain assumptions that may partially favor this interpretation, the global cumulative hyperlink position chart previously mentioned, which appears in figure 15, reveals a tendency for users to select hyperlinks appearing earlier in an article.

Finally, as per the initial requirements:

- The system shall collect contextual information that may influence user navigation behavior, such as article metadata.

This requirement is fulfilled, and can be validated via the bar chart visualization tied to the node selection tool.

- The system shall correlate navigation paths with contextual and environmental variables to identify potential influencing factors.

While correlation with the mentioned factors may be difficult to prove, based on the necessary assumptions in this project, the conjunction of the bar chart tool, the graph node selection tool, and the global cumu-

lative hyperlink position chart make correlations easier to visualize.

- The system shall support hypothesis-based analysis, acknowledging that causal inference may not be directly observable from the available data.

The various selection mechanisms and filtering tools, allow for very targeted hypothesis resolution.

- The system shall identify patterns that suggest potential motivations behind user choices, without assuming definitive causality.

This requirement is fulfilled both via the bar chart and global cumulative hyperlink position chart, and the Sankey diagram tool, in which intermediate node clusters can be easily spotted, such as the chronological jumps mentioned previously in this subsection.

6 Applications

Even though this work has been focused around the Wikispeedia database, the fact remains that local web navigation analysis is a task that many people may benefit from, for instance in the fields of marketing and sales analysis, the parallel can be made, in which given the home page of the store h and a desired product p a game $\mathbf{G}$ may be defined by $(s, t) = h, p$. With this setup, the main efforts of this work can be easily re-targeted, since navigation data is a thing most web pages already collect, via Google analytics or similar platforms.

It is also worth mentioning, this platform can be extended beyond web navigation, it can be extended to real navigation, and any problem that can be abstracted to graph navigation. For instance, by subdividing the earth into sectors, and marking one sector as origin and another as target, the migratory patterns of birds could be studied as a graph, and thus this tool could be used to analyze the tracks. This works as an example of the generalization of graph navigation analysis, but this can be applied to web shopping, to analyze why a product may not be getting visibility, navigation within a shop to certain products, and more.

7 Sustainability report and Ethical implications

The objective of this thesis is the design and development of a tool for the analysis of directed graph navigation without *teleportation*. Although the project does not directly involve physical resource consumption, its sustainability implications can be considered from the perspective of software engineering, computational efficiency, and long-term usability.

The developed tool contributes to sustainable software practices by focusing on efficient graph traversal and navigation methods. By saving resources in the cache, and writing graph information to disk, energy consumption implications are reduced, and can be shared to avoid recomputation within large groups of people.

From a social perspective, tools that improve the exploration and understanding of complex networks can support research and decision-making in various domains, such as knowledge representation, infrastructure analysis, and information systems. Providing a transparent navigation mechanism can also contribute to better interpretability of graph-based analyses.

Regarding ethical implications, it must be mentioned that when used for morally gray, or outright wrong objectives, navigation analysis can be dangerous. The most clear instance would probably be social networks, for the movement of users within social networks can be abstracted to graph navigation, and by focusing on maximizing revenue, or time on the platform, this can lead to psychological problems in users, that companies may ignore for the sake of profit. However, the main focus of this tool is to analyze concrete paths, and for this reason, social media navigation, which would have each user have a different starting path, would be less than ideal scenarios to employ it.

8 Conclusions

In conclusion, through the analysis of local web navigation without *teleportation*, all three original research questions have been addressed. These questions have not only been individually answered, but also collectively generalized through the development of a dedicated analytical tool designed to study the proposed problem space in a systematic manner.

The usefulness and effectiveness of this tool are reflected both in the extent to which it enables the resolution of the research questions and in its ability to satisfy the previously defined requirements. However, its value extends beyond this structured evaluation. It is further enhanced by its operational flexibility, which allows for interactive and exploratory analysis of navigation data.

This operability makes it possible to formulate and investigate additional ad hoc questions beyond the initial scope of the study, thereby providing a more comprehensive and adaptable framework for understanding user navigation behavior. This framework can also be extended for other navigation visualizations, such as the scenarios mentioned in the applications section.

8.1 Future work

Although the objectives proposed by this work are complete to a degree, future work remains as extending the tool to properly fulfill the intended motivation. For this reason, getting significative data of games $\mathbf{G}$ defined by $(s, n_{i..j})$ and extending the tool to visualize environments with one objective and multiple targets, to properly understand positioning between targets, and thus within domain sorting methods.

Special thanks

Finally, I wish to acknowledge the support of my family and friends, as well as from my tutor who have helped me in my work throughout this thesis. Special thanks are due to my wife and her family for their contribution as the principal testers in the local experimental studies.

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Appendices

A Data structure

The files used to generate the dataset where:

- paths_finished.tsv
- articles.tsv
- shortest-path-distance-matrix.txt

The files contain format specifications at their header, specifically, the first file "paths_finished.tsv":

```
1 # Successful (i.e., finished) Wikispeedia paths.
2 # Article names are URL-encoded; e.g., in Java they can be
   decoded using java.net.URLDecoder.decode(articleName, "UTF
   -8").
3 # Articles in a path are separated by ";".
4 # Back clicks are represented as "<".
5 # Ratings are optionally given by the user after finishing
   the game and range from 1 ("easy") to 5 ("brutal").
6 # Missing ratings are represented as "NULL".
7 # FORMAT: hashedIpAddress timestamp durationInSec path
   rating
8 #
9 # When publishing on this data set, please cite:
10 # (1) Robert West and Jure Leskovec:
11 #   Human Wayfinding in Information Networks.
12 #   21st International World Wide Web Conference (WWW), 2012.
13 # (2) Robert West, Joelle Pineau, and Doina Precup:
14 #   Wikispeedia: An Online Game for Inferring Semantic
   Distances between Concepts.
15 #   21st International Joint Conference on Artificial
   Intelligence (IJCAI), 2009.
16
17 6a3701d319fc3754 1297740409 166 14th_century;15th_century
   ;16th_century;Pacific_Ocean;Atlantic_Ocean;Accra;Africa;
   Atlantic_slave_trade;African_slave_trade NULL
18 ...
```

The second file `articles.tsv`:

```
1 # The list of all articles.
2 # Article names are URL-encoded; e.g., in Java they can be
   decoded using java.net.URLDecoder.decode(articleName, "UTF
   -8").
3 # FORMAT:  article
4 #
5 # When publishing on this data set, please cite:
6 # (1) Robert West and Jure Leskovec:
7 #   Human Wayfinding in Information Networks.
8 #   21st International World Wide Web Conference (WWW), 2012.
9 # (2) Robert West, Joelle Pineau, and Doina Precup:
10 #   Wikispeedia: An Online Game for Inferring Semantic
   Distances between Concepts.
11 #   21st International Joint Conference on Artificial
   Intelligence (IJCAI), 2009.
12
13 %C3%81ed%C3%A1n_mac_Gabr%C3%A1in
14 ...
```

And the third file `shortest-path-distance-matrix.txt`:

```
1 # The shortest-path distances between all pairs of articles,
   computed using the Floyd-Warshall algorithm.
2 # FORMAT:
3 # One row per article (the "source" of the shortest paths),
   in the same order as in articles.tsv.
4 # Each row contains the distances from the source to all
   articles (the "targets" of the shortest paths),
5 # again in the order of articles.tsv.
6 # The shortest-path distance is represented as a single digit
   , with no separators between values. This
7 # is possible because the longest shortest path happens to be
   of length 9.
8 # An underscore ("_") is used to indicate that the target
   cannot be reached from the source.
9 #
10 # When publishing on this data set, please cite:
11 # (1) Robert West and Jure Leskovec:
12 #   Human Wayfinding in Information Networks.
13 #   21st International World Wide Web Conference (WWW), 2012.
14 # (2) Robert West, Joelle Pineau, and Doina Precup:
15 #   Wikispeedia: An Online Game for Inferring Semantic
   Distances between Concepts.
16 #   21st International Joint Conference on Artificial
```



```

17 Intelligence (IJCAI), 2009.
18 0_____33333325634333435_2433544334_3_4...
19 ...

```

And the program generates, from a game $\mathbf{G}$ defined by (s, t) a file as $\langle s \rangle$
 $-- \langle t \rangle .csv$ containing edge data, and html article data, which allows for
the generation of the charts. Specifically the columns it contains are:

- Origin node
- Target node
- Position in html list
- number of uses of the edge

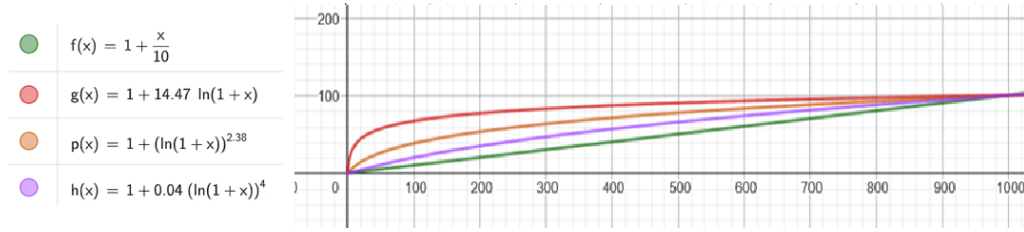


Figure 22: Comparison of equations scaling size

B Size scaling

Due to the high dispersion of the collected data, as illustrated by the various charts presented in Section 4.5, a linear scale proved inadequate for its visual representation. This issue can be observed in figure 15, which compares several candidate scaling functions. The selection of an appropriate scaling method was guided by the observation that the most frequently visited node accumulated approximately 1000 visits. Consequently, candidate functions were normalized such that they reached a value of ($y = 100$) at ($x = 1000$), providing a common reference point for visual size regulation.

The first alternative considered was a logarithmic function, a commonly employed approach for representing highly skewed data distributions. However, the resulting visualization exhibited excessive size compression, causing many differences between nodes to become visually indistinguishable. To mitigate this effect while preserving the desirable properties of logarithmic scaling, a logarithmic function raised to the second power was evaluated. Although this modification improved the visual differentiation between values, it did not sufficiently address the original issue. As a result, further adjustments were performed, leading to the final scaling function shown in figure 22, where the considered alternatives are compared.

The final function adopted for the visualization is $1 + 0.04 \ln(1+x)^4$. This expression can be interpreted as a scaled fourth-power logarithmic transformation, selected empirically to provide a more balanced distribution of node sizes while maintaining an upper value of approximately 100 for the maximum observed visit count.

C Observed results among games

During the development of the work, various different games $\mathbf{G}$ have been evaluated, some of the resulting visualizations are added here. This appendix also displays specific cases, as edges with progress -2 in figure 23, or examples of a sink node, in figure 24.



Figure 23: Graph of game **G** defined by $(s, t) = (\text{Bird}, \text{Great White Shark})$, unfiltered, node Orca selected, with an edge of progress -2; (Orca, Fish)



Figure 24: Graph of game **G** defined by $(s, t) = (\text{Bird}, \text{Adolf Hitler})$, with maximum filter and node Germany selected.

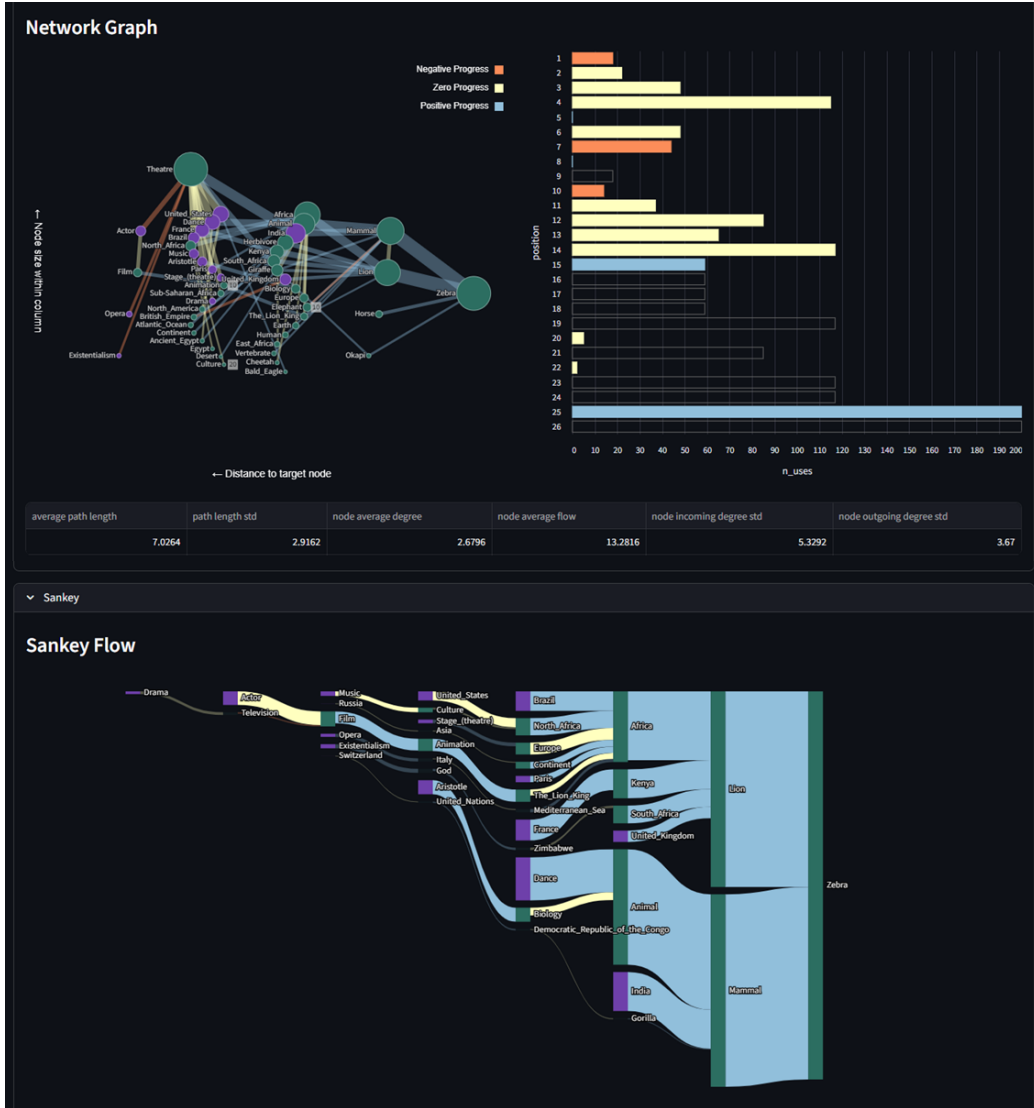


Figure 25: Graph and Sankey diagram of game G defined by $(s, t) = (\text{"Theatre"}, \text{"Zebra"})$ with an edge filter of 5% and all Sankey diagram flows visualized, with labels on



Figure 26: Graph and Sankey diagram of game $\mathbf{G}$ defined by $(s, t) = (\text{Bird}, \text{Great White Shark})$ with an edge filter of 5% and half Sankey diagram flows visualized, where the graph draws semblance to a fish, which happens to be the name of the most visited node.

D Local experiments

During the development of the project, the generated data and preliminary findings were informally contrasted through a series of small-scale local experiments. These experiments were not designed as a formal user study and therefore do not possess statistical validity, primarily due to the limited number of participants and the absence of a rigorous experimental protocol, including institutional ethics review, standardized procedures, and comprehensive documentation.

Rather than constituting a structured evaluation, these activities should be understood as exploratory observations conducted during the development process. A small group of volunteer participants, recruited from my personal network, were invited to play a set of three game sessions. Prior to participation, all individuals were informed about the purpose of the activity, the nature of the data being collected, and the intended use of the recorded information.

During gameplay, player movement and interaction traces were recorded for each session. These datasets were subsequently incorporated on a temporary basis into the developed visualization tool to assess its functionality and to qualitatively inspect the resulting visual representations. The observations derived from these exploratory trials served primarily as an informal validation mechanism during development and should not be interpreted as empirical evidence supporting the effectiveness or generalizability of the proposed approach.

Some of the results can be seen in figures 27, 28 and 29, where within the game **G** three players end in the same flow (Agriculture, Peas, Beans), which happens to be part of the main flow.

Some more exploratory attempts were also recorded, such as the one seen in figure 30

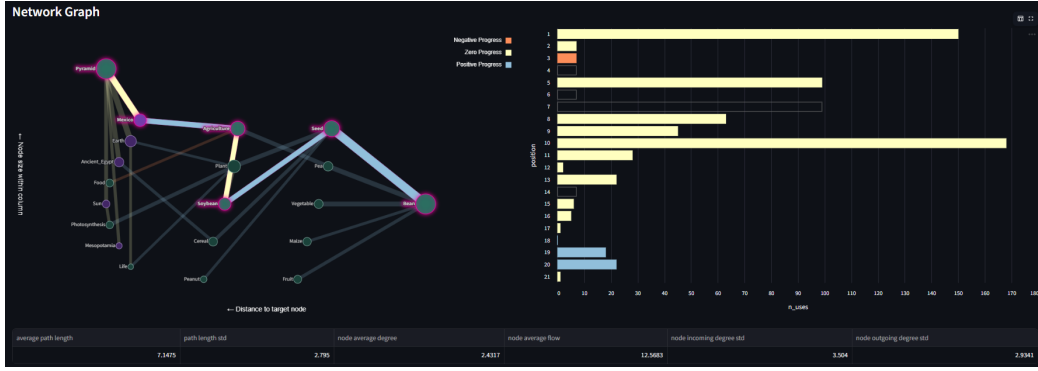


Figure 27: Track of game $\mathbf{G}$ defined by $(s, t) = (\text{Pyramid}, \text{Bean})$, with an edge filter of 5% by player A

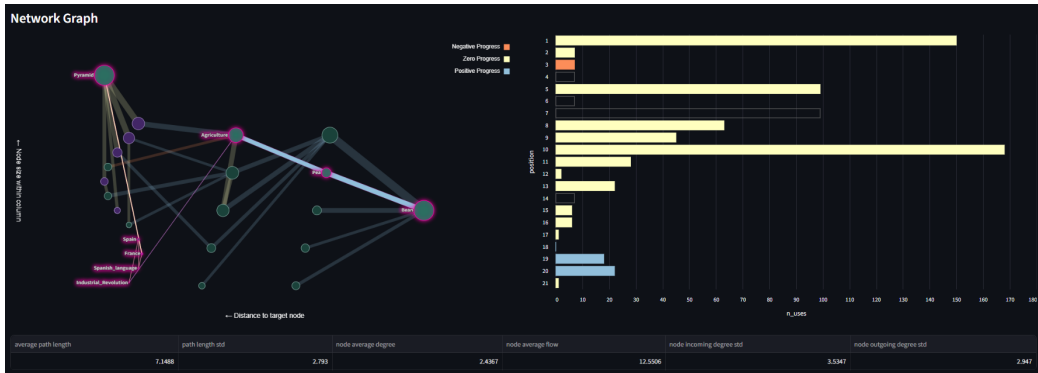


Figure 28: Track of game $\mathbf{G}$ defined by $(s, t) = (\text{Pyramid}, \text{Bean})$, with an edge filter of 5% by player B

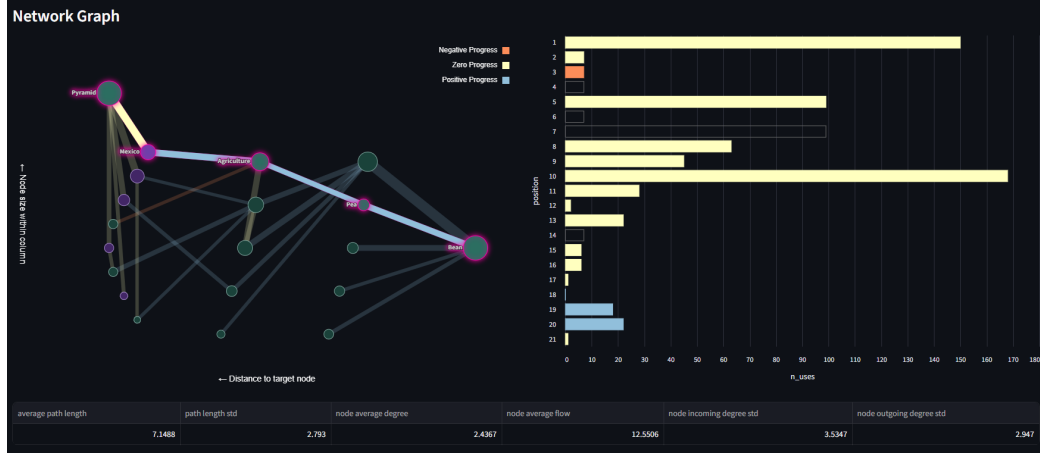


Figure 29: Track of game $\mathbf{G}$ defined by $(s, t) = (\text{Pyramid}, \text{Bean})$, with an edge filter of 5% by player C

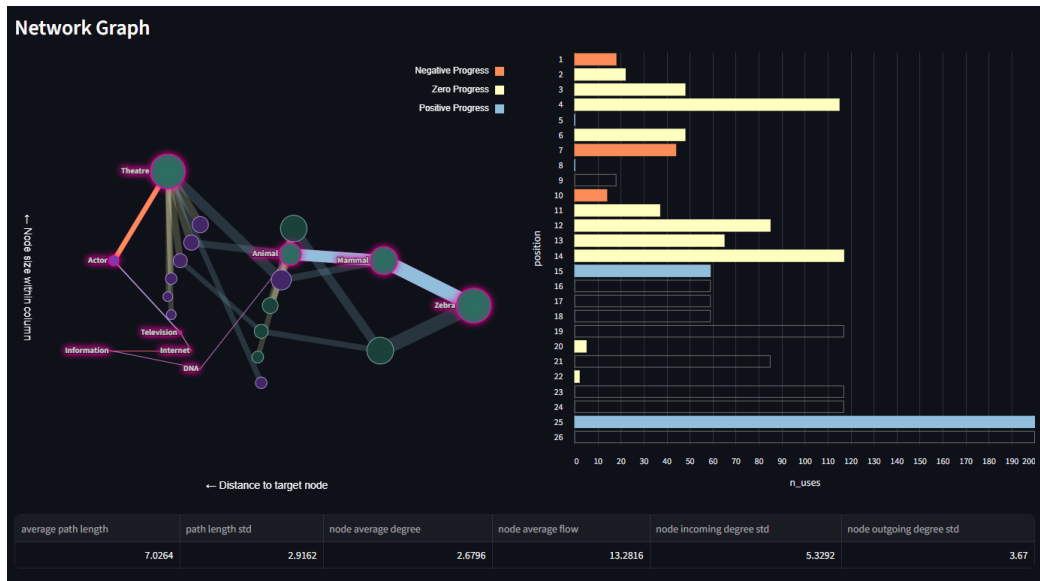


Figure 30: Track of game $\mathbf{G}$ defined by $(s, t) = (\text{"Theatre"}, \text{"Zebra"})$, with an edge filter of 5% by player B, in which the path is over double the minimum distance